End-sem: worked solutions

22 Nov 2024 · 40 marks · 3 hours (open code book) · CEPC13

Q1 (a)

2 marks

> Arrange the stages in order for the wet process of cement manufacture: raw meal silo, rotary kiln, slurry preparation by water mixing, slurry preheater, ball mill, gypsum addition, limestone and shale crushing, packing and distribution, cement silo.

1. **Limestone and shale crushing**
2. **Slurry preparation by water mixing** (wash mill / wet grinding)
3. **Raw meal silo** (slurry storage and correction tanks)
4. **Slurry preheater**
5. **Rotary kiln** → clinker (then cooled)
6. **Gypsum addition**
7. **Ball mill** (grinding clinker with gypsum)
8. **Cement silo**
9. **Packing and distribution**

Q1 (b)

2 marks

> Discuss (with equations where relevant): "Gypsum is an essential ingredient in cement manufacture. It, however, is also encountered as an outcome of concrete sulphate attack."

Gypsum as an essential ingredient. Without it, C_3A reacts instantly with water (**flash set**). Gypsum (3–5%) reacts with C_3A to form **ettringite**, which coats the C_3A grains and slows their hydration, giving a workable dormant period:

$$C_3A + 3C\bar{S}H_2 + 26H \rightarrow C_6A\bar{S}_3H_{32} \text{ (ettringite)}$$

Later, once gypsum is consumed: $2C_3A + C_6A\bar{S}_3H_{32} + 4H \rightarrow 3C_4A\bar{S}H_{12}$ (monosulphate).

Gypsum as a product of sulfate attack. External sulfates (soil, groundwater, sea water) react with the CH in hardened concrete to form gypsum, with expansion and softening:

$$Ca(OH)_2 + Na_2SO_4 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O + 2NaOH$$

The gypsum then reacts with monosulphate (and remaining aluminates) to form **secondary ettringite**, which is highly expansive and cracks the hardened concrete:

$$C_4A\bar{S}H_{12} + 2C\bar{S}H_2 + 16H \rightarrow C_6A\bar{S}_3H_{32}$$

With **magnesium sulfate** the attack is worse, because it also decomposes C-S-H:

$$Ca(OH)_2 + MgSO_4 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O + Mg(OH)_2$$

So the same compound is beneficial when added in the right amount at the cement plant, but harmful when it forms inside hardened concrete where there is no room for the expansion.

Q1 (c)

2 marks

> Write the chemical equations for (i) formation of C_4AF from raw ingredients of cement manufacture, (ii) leaching of concrete.

(i) $4CaCO_3 \rightarrow 4CaO + 4CO_2$ (calcination); then in the kiln melt:

$$4CaO + Al_2O_3 + Fe_2O_3 \rightarrow 4CaO \cdot Al_2O_3 \cdot Fe_2O_3 \ (C_4AF)$$

(ii) Leaching: soft or acidic water dissolves calcium hydroxide, which then reacts with CO_2 to deposit white calcium carbonate on the surface (efflorescence):

$$Ca(OH)_2 \text{ (dissolved)} + CO_2 \rightarrow CaCO_3 \downarrow + H_2O$$

With CO_2-rich (aggressive) water the carbonate itself dissolves: $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$ (soluble), so the paste keeps losing lime and becomes porous.

Q2 (a)

4 marks

> Identify the tests depicted by the images. Also state what each test is used to find.

Image	Apparatus / test	Used to find
(i) Small split brass cylinder with two long pointers and glass plates	**Le Chatelier mould: soundness test** (IS 4031 Part 3)	**Unsoundness of cement due to free lime**: expansion between the pointer tips after boiling (limit 10 mm)
(ii) Beam on two support rollers, loaded through two rollers from above	**Flexural strength test (third-point loading)** (IS 516)	**Modulus of rupture** of concrete: $f_b = PL/bd^2$
(iii) Enclosed drum cabinet with revolution counter and switches	**Los Angeles abrasion testing machine** (IS 2386 Part 4)	**Abrasion resistance (hardness) of coarse aggregate**: % wear after the specified revolutions with steel balls
(iv) Frame with a dial gauge on top and a specimen bar between the posts	**Length comparator**	**Length change** of bars: autoclave soundness (expansion < 0.8%), drying shrinkage, and mortar-bar alkali–aggregate expansion

(Image (iii) is unclear in the scan; the cabinet shape with a drum housing and counter matches an enclosed LA abrasion machine.)

Q2 (b)

2 marks

> What does the number XX stand for in (i) OPC XX for cement, (ii) M XX for concrete? How are these quantities defined?

- **OPC XX** (e.g. OPC 43): XX is the **grade of cement = minimum 28-day compressive strength in MPa** of **1 : 3 cement : standard sand mortar cubes (70.6 mm, area 5000 mm²)**, w = (P/4 + 3)% of the dry mass, vibrated and moist cured (IS 4031 Part 6; IS 269).
- **M XX** (e.g. M35): **M = mix**; XX is the **characteristic compressive strength in MPa** of **150 mm concrete cubes at 28 days**, defined as the strength **below which not more than 5% of the test results are expected to fall** (IS 456). $f_{ck} = f_{mean} - 1.65\ s$.

Q2 (c)

4 marks

> Perform an M35 mix design using IS 10262:2019: cement OPC 43; maximum nominal aggregate size 12 mm (sub-angular); member: heavily reinforced footing in non-aggressive soil; specific gravities: cement 3.10, fine aggregate 2.68, coarse aggregate 2.74, superplasticizer (if used) 1.16; fine aggregate zone III; water absorption: fine aggregate 0.8%, coarse aggregate 0.5%. Take unspecified details from code books.

Assumptions (from codes):

- Footing in non-aggressive soil → **moderate exposure** (IS 456 Table 3): max w/c **0.50**, min cement **300 kg/m³**, min grade M25. IS 10262 tables cover 10 and 20 mm only, so values for **12.5 mm are interpolated**; min cement with the 10 mm adjustment (+40) = **340 kg/m³**.
- Heavily reinforced section → **medium workability, slump 100 mm** (IS 456 Cl. 7.1).
- Good supervision (s = 5 MPa); SP 1% of cement giving 20% water reduction; air 1%; aggregates dry (absorption added).

Step	Working	Result
Target strength	max(35 + 1.65 × 5, 35 + 6.5)	**43.25 MPa**
w/c	Fig. 1, OPC 43 curve → about 0.40 (< 0.50 ✓)	**0.40**
Water (50 mm slump)	208 (10 mm) − ¼(208 − 186) = 202.5 for 12.5 mm; −10 (sub-angular) = 192.5	192.5 kg
Slump 100 mm	+6% → 204.1; −20% with SP	**163.2 kg/m³**
Cement	163.2 / 0.40 (≥ 340, ≤ 450 ✓)	**408 kg/m³**
SP	1%	**4.1 kg/m³**
CA fraction (zone III)	0.48 (10 mm) to 0.64 (20 mm) → 0.52 at 12.5 mm; + 0.02 for w/c 0.40	**0.54**

Volume (m^3)	
Air	0.010
Cement 408 / 3100	0.1316
Water	0.1632
SP 4.1 / 1160	0.0035
All-in aggregate	**0.6916**
CA (SSD) = 0.6916 × 0.54 × 2740	**1023 kg**
FA (SSD) = 0.6916 × 0.46 × 2680	**853 kg**

Absorption (dry aggregates): CA 1023/1.005 = **1018 kg**, FA 853/1.008 = **846 kg**; extra water 11.9 kg → **water to add 175 kg**.

Trial mix per m^3: OPC 43 **408 kg**, water **175 kg** (free w/c 0.40), FA **846 kg**, CA 12.5 mm **1018 kg**, SP **4.1 kg**. Ratio (SSD) **1 : 2.09 : 2.51**.

Q3

6 marks

> Fill in the blanks:
>
> (i) The main 4 final compounds in cement powder are collectively called ____.
>
> (ii) The second highest raw material (compound name and formula) in cement manufacture is ____.
>
> (iii) The sieve size used to find the mass lost by powdering during the crushing test and impact test of coarse aggregates is ____ mm.
>
> (iv) The maximum cement content allowed in a concrete mix by IS 456:2000 when fly ash or GGBS are not added is ____ kg per cubic metre.
>
> (v) UPV stands for ____.
>
> (vi) In corrosion of embedded steel, the rusted area acts as the ____ (cathode/anode) and the adjacent areas act as the ____ (cathode/anode).

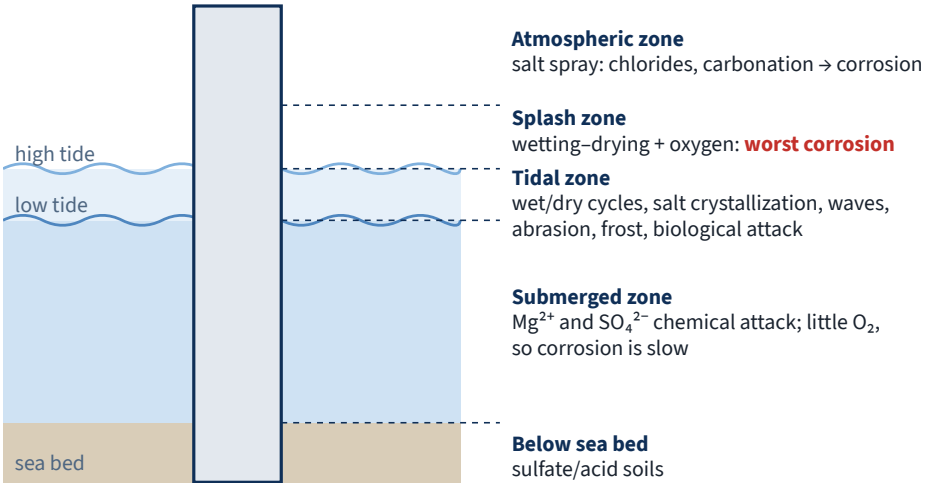
1. **Bogue compounds** (clinker minerals: C_3S, C_2S, C_3A, C_4AF).
2. **Silica, silicon dioxide (SiO_2)**, supplied by clay/shale (17–25% of cement, second after lime, CaO from $CaCO_3$).

3. **2.36 mm**.
4. **450 kg/m³** (IS 456 Cl. 8.2.4.2).
5. **Ultrasonic Pulse Velocity**.
6. Rusted area = **anode**; adjacent areas = **cathode**.

Q4 (a)

2 marks

> With a diagram, explain how tidal variation affects concrete structures that are partly submerged and partly above sea water.



- **Atmospheric zone**: airborne salt spray deposits chlorides; carbonation; corrosion of reinforcement.
- **Splash zone** (just above high tide): repeated wetting and drying concentrates chlorides, and oxygen is freely available → **most severe reinforcement corrosion**, cracking and spalling.
- **Tidal zone** (between high and low tide): alternate wetting and drying, **salt crystallization** in pores, wave impact and **abrasion** by sand and floating objects, **freeze–thaw** in cold climates, marine growth.

- **Submerged zone**: always saturated, so oxygen is scarce and corrosion is slow, but **chemical attack** by Mg^{2+} and SO_4^{2-} (brucite and gypsum/ettringite formation) and leaching occur.

Protection: low w/c (≤ 0.40–0.45), slag/fly ash/silica fume, cover ≥ 50–75 mm, good curing, coatings in the splash zone.

Q4 (b)

2 marks

> Discuss Litvan's theory of concrete damage due to frost action.

Litvan (1972) explained frost damage by **moisture redistribution**, not by simple ice expansion:

- Water held in the **very small pores (gel and fine capillary pores)** is adsorbed and structured, so it **cannot freeze in place** at 0 °C; it becomes **supercooled**.
- Supercooled water has a **higher vapour pressure than ice**. This difference drives the water to **migrate out of the small pores** to places where it can freeze: the surface, cracks, or large pores and air voids.
- If the concrete is **nearly saturated, permeability is low or the path is long**, the water cannot leave fast enough. **Internal pressure** and drying (desiccation) stresses build up and **crack** the paste.
- **Entrained air voids**, closely spaced (spacing factor < 0.2 mm), give nearby empty sites for ice to form, shortening the migration path; this is why air entrainment protects concrete.

Q4 (c)

2 marks

> Compare the fresh concrete constituents and properties of high performance concrete and high density concrete.

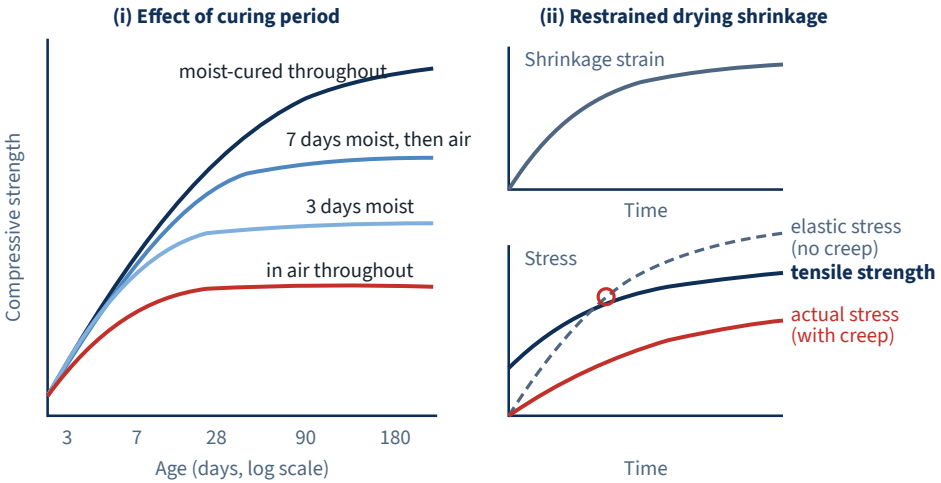
	High performance concrete (HPC)	High density (heavyweight) concrete
Purpose	High strength, durability, workability	**Radiation shielding** (nuclear plants, X-ray rooms), counterweights

	High performance concrete (HPC)	High density (heavyweight) concrete
Density	Normal (≈ 2400–2500 kg/m^3)	**3500–5500 kg/m^3**
Aggregates	Strong, well-graded, small (10–12.5 mm) normal aggregate	**Heavy aggregates**: barite (SG ≈ 4.2), magnetite/hematite (≈ 4.5–5), limonite, steel shot/punchings; boron-bearing materials for neutron shielding
Binder and water	**Very low w/b (0.25–0.35)**, high binder (450–600 kg/m^3), **silica fume, fly ash, GGBS**	Ordinary binder content and w/c; hydrous ores to hold water for neutron shielding
Admixtures	High-range **superplasticizer**, sometimes VMA	Plasticizers as needed
Fresh properties	**High flowability yet cohesive**, little bleeding, can be sticky, slump loss; prone to plastic shrinkage → early curing	**Prone to segregation** (heavy particles sink), low workability, needs careful placing (often **preplaced aggregate** method) and **strong formwork**

Q4 (d)

2 marks

> Draw graphs that depict (i) the effect of curing period on long-term compressive strength, and (ii) strain vs time and stress vs time during restrained drying shrinkage.



- **(i)** Continuously moist-cured concrete keeps gaining strength; the shorter the moist curing, the earlier the strength levels off; concrete never moist-cured reaches only about half the strength.
- **(ii)** Shrinkage strain rises quickly at first and then flattens. If the member is restrained, it develops **tensile stress**. Without creep, the elastic stress would cross the tensile strength (circle) and the concrete would crack early. **Creep relaxes** the stress, so the actual stress stays lower and cracking is delayed or avoided.

Q4 (e)

2 marks

> Differentiate between (i) compressive strength test of concrete using cubes and using cylinders, (ii) immersion vibrators and formwork vibrators.

(i)

Cubes	Cylinders
150 mm (100 mm) cubes; IS 516, BS	150 × 300 mm; ASTM, ACI, Eurocode
Tested **on the side** (smooth mould faces), no capping	Tested **on the end**; top must be **capped or ground**

Cubes	Cylinders
h/d = 1 → strong **platen restraint**, higher apparent strength	h/d = 2 → middle zone is free of restraint, nearer **true uniaxial strength**
Hourglass (double pyramid) failure	Cone/shear or splitting failure
Cylinder strength ≈ **0.8 × cube strength**	

(ii)

Immersion (internal, needle) vibrator	Formwork (external) vibrator
Poker head **inserted into the concrete**	**Clamped to the formwork**; vibrates form and concrete
Most efficient: energy goes straight into the concrete	Less efficient; much energy is spent vibrating the forms
Head 20–180 mm; high frequency	1000–5000 cycles/min
Mass concrete, beams, columns, slabs	**Thin sections, congested reinforcement, precast**, where pokers can't enter
Ordinary formwork	Forms must be **strong and tight** (no grout leakage)

Q5 (a)

4 marks

> Mention the steps that can be taken (with reasoning) for: (i) reducing alkali–aggregate reaction, (ii) transporting concrete across long distance and duration, (iii) concreting in regions of high temperature and (iv) reducing permeability of concrete without affecting workability in the fresh concrete stage.

(i) AAR: use non-reactive aggregate; low-alkali cement (Na_2O eq. < 0.6%), limit total alkali (< 3 kg/m^3); replace cement with **fly ash, GGBS, silica fume, metakaolin** (dilute and bind alkalis); lithium admixtures; keep the concrete dry (the gel swells only with water).

(ii) Long-haul transport: use **ready-mixed concrete in transit mixers / agitator trucks** (2–6 rpm) to prevent segregation; add a **retarder or retarding superplasticizer** to keep the slump; cool the mix in hot weather and cover the

drum; limit the time to discharge (about 90 min or 300 drum revolutions, ASTM C94); **do not add water at site** (re-dose SP instead); keep the mix cohesive (good grading).

(iii) Hot weather: cool ingredients (chilled water, ice, shaded aggregates), keep concrete below about 32 °C; **retarders**, low-heat or blended cement; place at night/cool hours and quickly; wet the forms; windbreaks, fog spray, curing compound and **immediate curing** to prevent plastic shrinkage cracks.

(iv) Lower permeability, same workability: reduce w/c with a **superplasticizer**; add **mineral admixtures** for pore refinement (fly ash also helps workability); good grading and compaction; **adequate moist curing**.

Q5 (b)

4 marks

> A 10-storey RC building is being built at a location prone to high water table and flooding, where temperatures can approach freezing. The bottom floor will house a high-pressure burner, requiring the floor to withstand 2 hours of fire in an accident. Beams and columns have an average clear spacing of 60 mm. Answer with reasoning: (i) can 40 mm aggregates be used? (ii) what admixtures are advisable? (iii) what slump value and conveyance method suit concreting up to 10 floors? (iv) are any special considerations required for the footing compared with slabs, beams and columns?

(i) 40 mm aggregate: not for beams and columns. The spacing rules alone (≤ clear spacing − 5 = 55 mm; ≤ ¾ × 60 = 45 mm) would allow 40 mm, but IS 456 Cl. 5.3.3 also limits size to **5 mm less than the minimum cover**: columns and beams have about 40 mm cover (more for 2-hour fire), so the limit is about 35 mm. 40 mm aggregate also **cannot be pumped** to 10 floors easily and is poor for fire spalling and freeze–thaw. Use **20 mm** for the frame; 40 mm may be used only in a massive footing.

(ii) Admixtures:

- **Superplasticizer**: low w/c for durability (water table, frost) with the flow needed for pumping.

- **Air-entraining admixture**: freeze–thaw resistance (near-freezing exposure).
- **Mineral admixtures** (fly ash/GGBS): low permeability against groundwater and flooding, lower heat in the footing.
- **Non-chloride accelerator** in cold weather (no $CaCl_2$: corrosion).
- **Integral waterproofing admixture** for the footing/basement.
- For the burner floor: **polypropylene fibres** to reduce explosive spalling in fire; avoid very high silica fume dosage there.

(iii) Slump and conveyance: pump (boom pump, or line pump with a vertical pipeline) to reach about 30–35 m; slump **75–100 mm** (IS 456 for pumped concrete), or 100–150 mm with superplasticizer to reduce pumping pressure; cohesive mix with adequate fines. (A tower crane with bucket is an alternative, but slower.)

(iv) Footing:

- **Dewatering** during excavation and concreting; place by **tremie** if water cannot be removed.
- Exposure class from groundwater chemistry (sulfates/chlorides): **low w/c (≤ 0.45), higher cement content, blended cement**, larger **cover (50–75 mm)**; waterproofing membrane.
- Check **uplift/buoyancy** from the high water table.
- Place below the **frost line**; protect fresh concrete from freezing.
- Massive footings: control **heat of hydration** (low-heat/blended cement, possibly larger aggregate).

Also, for **2-hour fire resistance** on the burner floor, provide IS 456 Table 16A covers (e.g. 40 mm for columns and simply supported beams) and minimum member sizes, and prefer calcareous or lightweight aggregate.